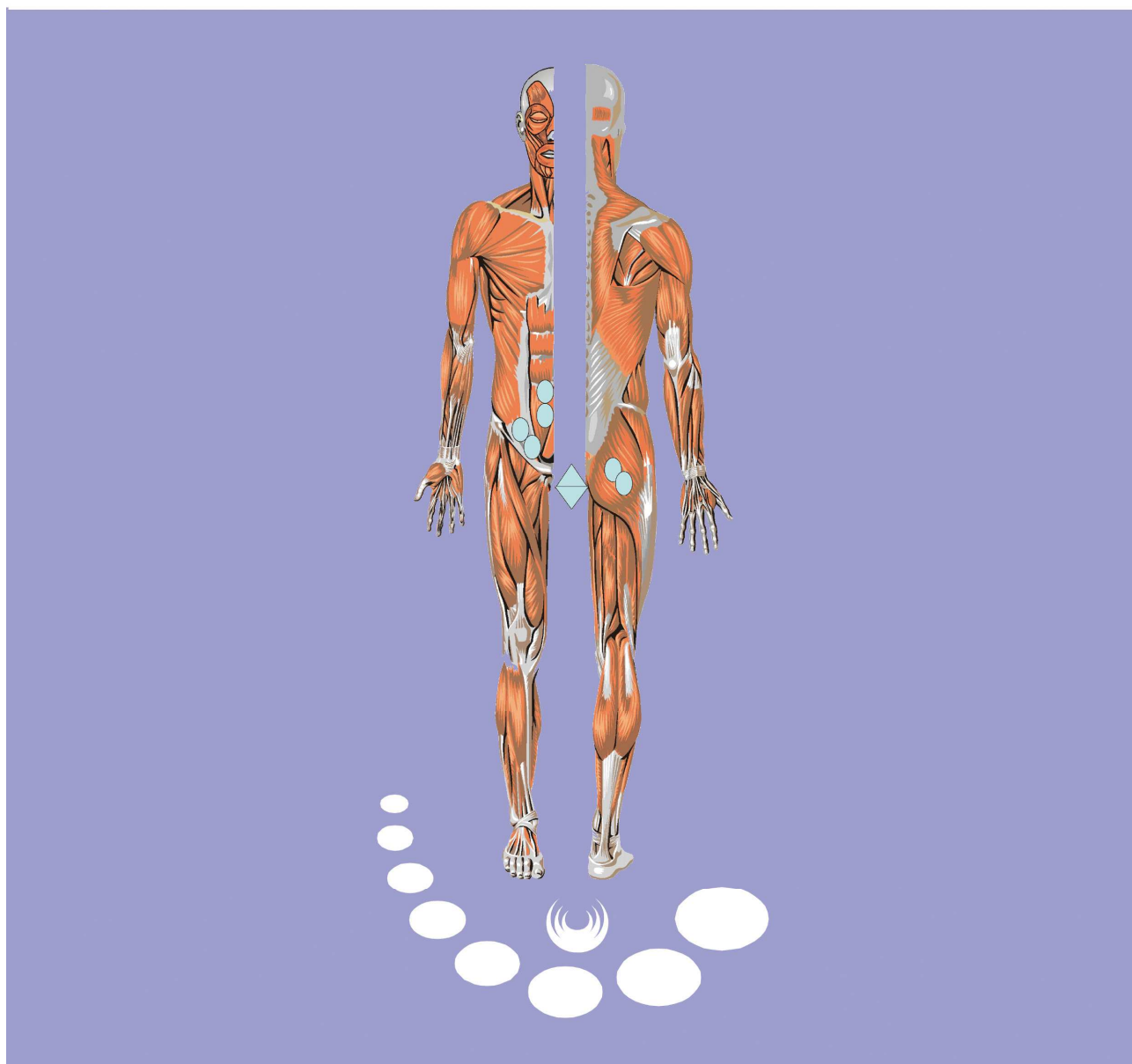


EMG-Based Evaluation & Therapy Concepts For Pelvic Floor Dysfunctions

Birgit Schulte-Frei und Dr. Peter Konrad

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EMG Based Evaluation & Therapy Concept for Pelvic Floor Dysfunctions

**Birgit Schulte-Frei
Peter Konrad**

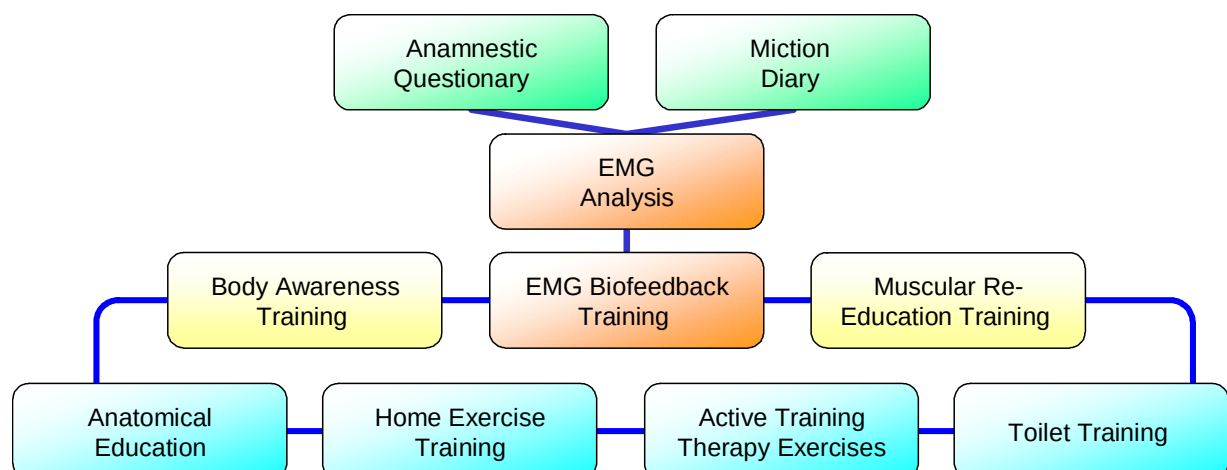
ProPhysio Rehabilitation Clinic
Cologne Germany

Introduction

Electromyography (EMG) is a well established method to directly measure the pelvic floor muscle innervation and then use this information for the analysis, documentation and training of pelvic floor dysfunctions (Biofeedback book). One and two channel measurements are commonly used setups to plan and perform treatment regimes for pelvic floor dysfunctions like fecal and urinal incontinence. The goal of our concept is to improve the established routines in terms of more accurate neuromuscular evaluation and more effective treatment modalities.

Conceptual Background

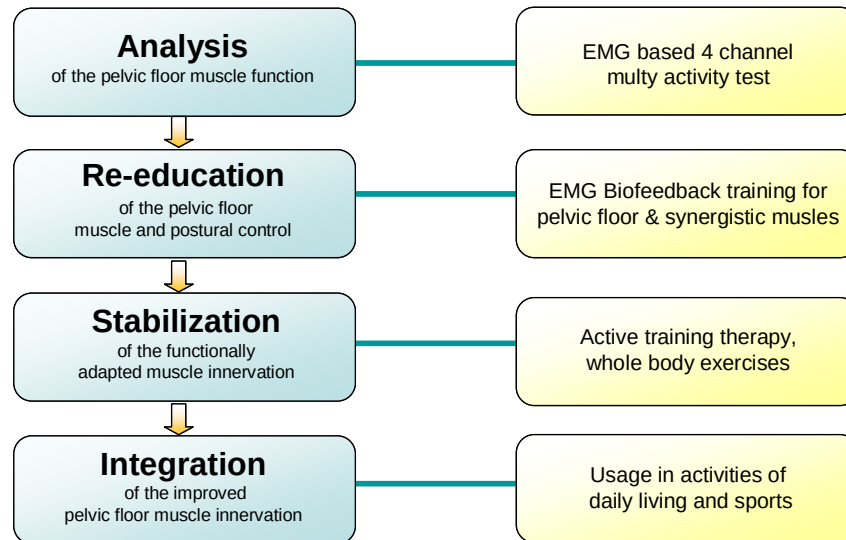
Our rehabilitation center is equipped with numerous biomechanical evaluation tools, medical strength training machines and cardio ergometers. Several modules are available within our concepts for treatment of pelvic floor dysfunctions:



The main modules are the EMG-based analysis and EMG biofeedback training. They are the main scope of this document.

The role of EMG as an evaluation and treatment tool

Our treatment concept is based on the analysis of the pelvic floor muscle as well as the surrounding muscles. The following graph overviews all stages within the concept:



Insurance companies in Germany pay for 12 sessions of pelvic floor training for each patient. One session takes 30 minutes. After the evaluation and EMG analysis procedures, we first concentrate on the isolated muscle function: relaxation or facilitation of the pelvic floor. The PT assisted lessons are performed once a week. The patient needs to comply with therapy by doing daily home exercise training. During later stages of the therapy feedback, controlled pelvic floor contraction exercises are combined with regular training therapy exercises and functional movements of daily activities. Multi-channel EMG serves as an effective control measure to activate deconditioned pelvic contractions, facilitated by dedicated use of synergistic muscles and increase the quality by detraining hyperactive global muscles. The patient also benefits from the general conditioning effect of training exercises.

In the following chapters each therapy stage is introduced in more detail.

1. Analysis of the pelvic floor muscle function

Based on previous work published by Shelly et al., Glazer, and Trautmann (literature citation will follow soon), we have established a standardized multi-EMG test setup using a surface EMG system manufactured by Noraxon USA, INC. (MyoSystem 1400A).

The 4 EMG channel approach

To enable a qualified detection of the neuromuscular coordination of the pelvic floor muscles, at least four EMG channels are needed. Besides the detection of the pelvic floor contraction itself, the activity of the synergistic (e.g. Internal oblique) muscle is important to measure. Furthermore, it is valuable to detect global co-contracting muscles (Gluteus max., Rectus abd.) that may mimic the pelvic floor contraction. Later, this knowledge is important for the coordination and isolation training ("Muscular Re-education") of the pelvic floor muscles. It teaches the patient how to contract the correct muscles during exercises and daily life activities.

At the first patient visit, we perform a sequence of established pelvic floor contraction activities. Using surface EMG, we measure the Gluteus Maximus, the Internal Obliques and the Rectus Abdominis. The pelvic floor muscles are detected by anal or vaginal probes (MEDICHECK-Germany).

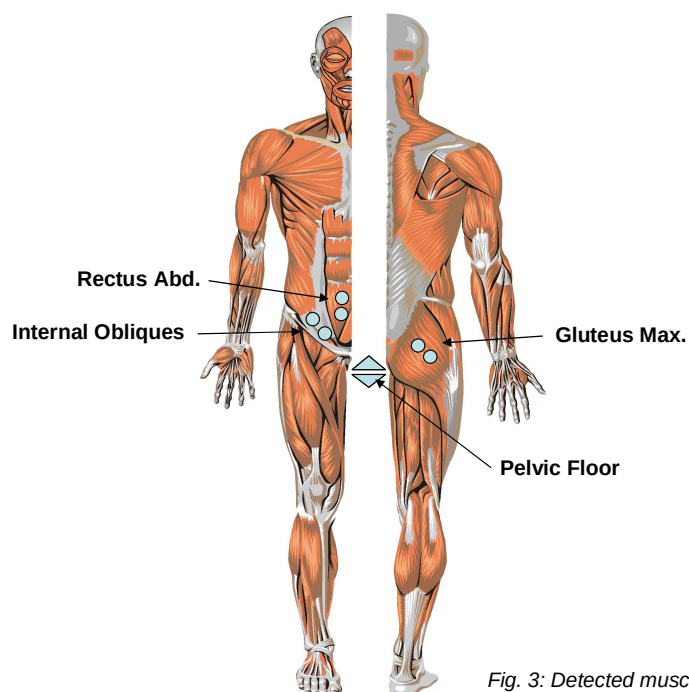


Fig. 3: Detected muscle sites



Figure 4: Anal (left) and vaginal (right) EMG probe

We measure the smoothed, rectified (RMS 100 ms) EMG signal in a bandwidth of 20 to 500 Hz. and at 1000 Hz sampling frequency. All data are acquired and analyzed using the clinical application protocol “Incontinence Multi-Activity Test” in **MyoResearch XP** (Noraxon USA, INC.).

After the electrode application, the patient performs a standardized sequence of pelvic floor activities, as proposed by Glazer:

EMG Baseline	5 seconds of muscle relaxation
Quick Flicks	5 fast upwards contractions and immediate relaxation
Maximal Up-Contractions	5 maximal contractions with 10 seconds duration/pausing
Endurance Hold	Static contraction of 30 – 60 sec. duration
Resting Tone	Immediate relaxation right after the Endurance Hold

The software automatically guides the user through the test sequence by prompting visual and acoustic contraction commands.

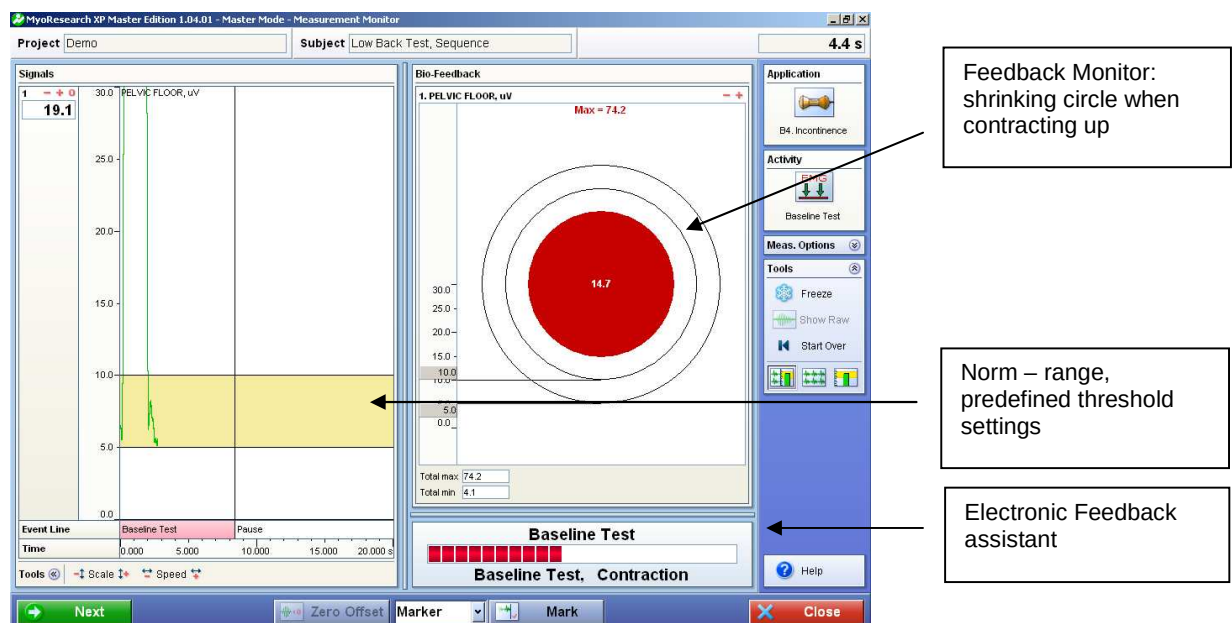


Fig. 5: Measurement & Biofeedback monitor of MyoResearch XP

All signals can be observed in real time; stored records are shown in the record viewer and the test results are analyzed in an automatic analysis report:

Record Viewer Screen:

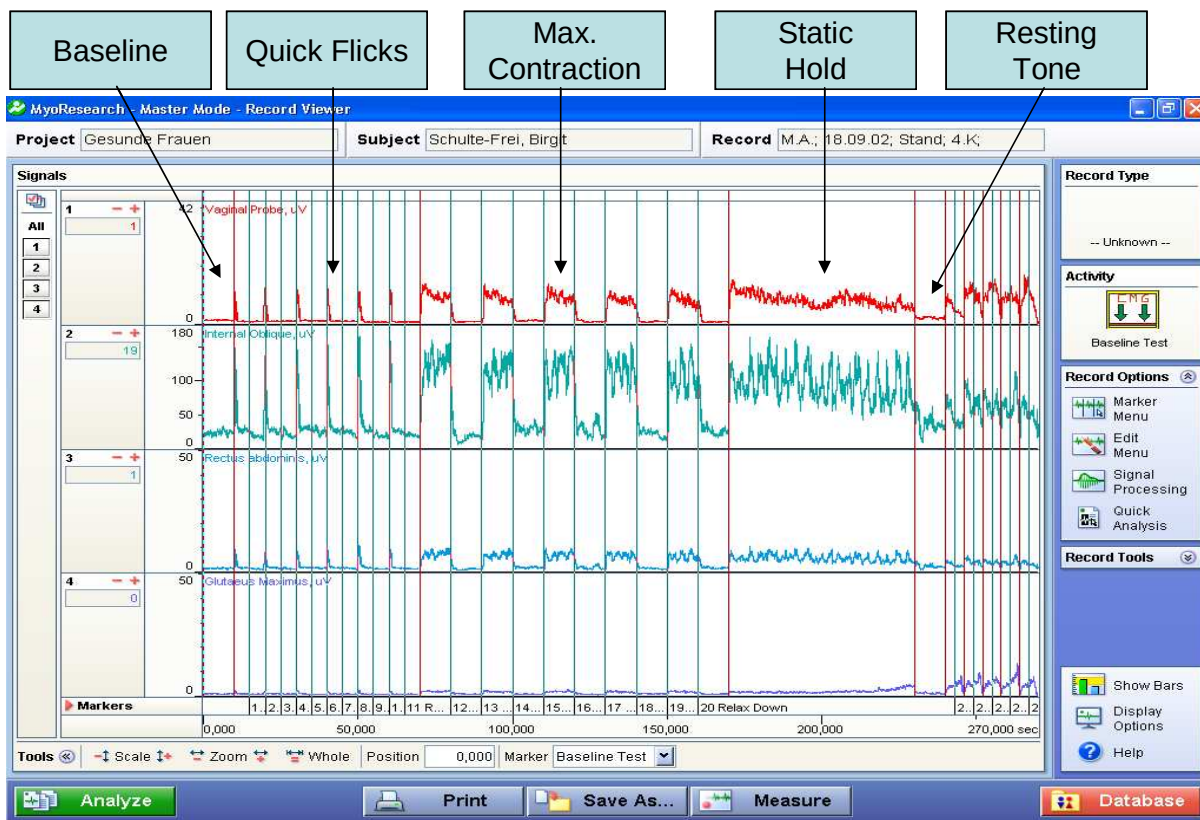
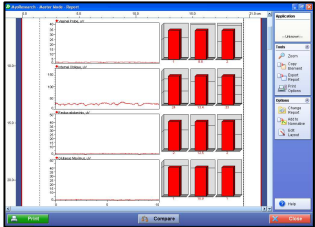
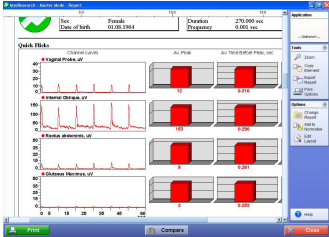
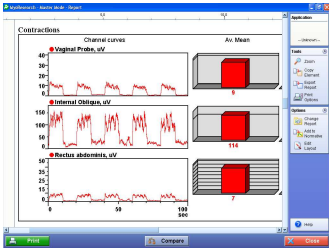
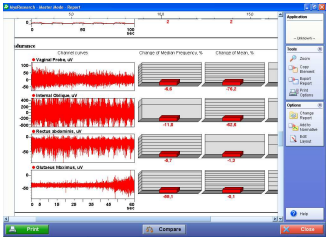
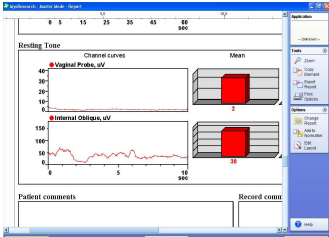


Fig. 6: Record Viewer showing all activities of the multi-activity test protocol in MyoResearch XP

The test report analyzes each activity with a set of individual parameters.

Activity:	Analysis Parameters and Findings
Baseline: 	Parameters: Mean EMG Amplitude, Coefficient of Variance Observation: No relaxation of the pelvic floor muscle in lying position Reduced activation in standing position Reduced co-activation of the M. oblique internus Diagnosis: Hypertonus in lying position Hypotonus in standing position
Quick Flicks: 	Parameters: Averaged Peak Amplitude, Time to/after Peak, Findings: slow peak increase during quick-flick bursts slow relaxation after quick activation reduced peak activation level co-activation of the surrounding muscles: M. gluteus, M. internus abdom. Reduced co-activation of the M. oblique internus Diagnosis: Muscular dysfunction, relaxation deficits
Maximal contraction 	Parameters: Averaged Mean Value Findings: reduced pelvic floor activation level steep decrease of activity within 10 seconds problem to innervate over 10 seconds co-activation of the M- transverses abdominis, M. gluteus max. Diagnosis: Muscular dysfunction, weakness, relaxation deficits
Endurance Hold 	Parameters: Amplitude and Frequency change over time Findings: Time domain changes due to fatigue Constancy of contraction level co-activation of M. gluteus max, M. rectus abdom. Diagnosis: Reduced endurance, innervation deficits
Resting Tone 	Parameters: Mean EMG Amplitude, Coefficient of Variance Findings: Increased rest line activity, late rest line level Diagnosis: Hypertonus, Relaxation Deficits

2 – Re-education of the pelvic floor muscle and postural control

The re-education program has two basic areas:

Up-/Downtraining to address hyper- or hypo-activity

“Uptraining” applies to weak or hypotonic pelvic floor muscles, “Downtraining” to hyperactive pelvic floor muscles. Based on the analysis of the pelvic floor EMG and the daily MVC (patients maximal voluntary contraction), the EMG feedback assisted therapy concentrates on muscular activation training.

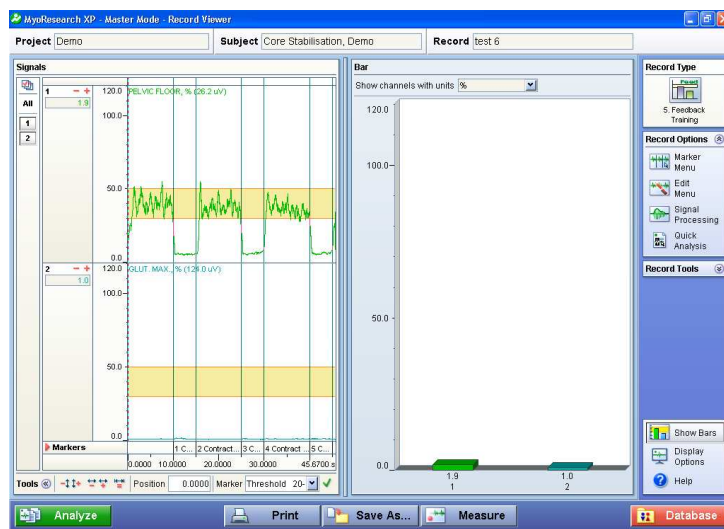


Fig. 7: Biofeedback recording with predefined threshold range (yellow area) for MVC normalized EMG contractions. Upper trace: pelvic floor, lower trace: gluteus max.

Isolated pelvic floor contraction to address coordinative deficits

By using at least 2 EMG channels, the patient can be trained to properly contract the pelvic floor muscle and sense the underlying body behavior. The focus here is the quality of isolation, not the amplitude. Hyperactive global muscles may need to be retrained to relax while performing the isolated pelvic floor contraction.



Fig. 8: Multi-channel EMG recording (left panel) with synchronized digital video (upper right picture) and real time analysis (lower right bar graphs). During this abdominal drawing exercise, high EMG is achieved for pelvic floor and internal oblique (ch. 1-2), while keeping the activation of global muscles low (ch. 3-4)

During the first 3 therapy sessions, the patient needs to “find” and improve his/her pelvic floor innervation. Later, this contraction needs to be coordinated with other synergistic muscles, i.e. the m. oblique internus. If the patient experiences difficulties contracting the target muscle, backward facilitation over the internal obliques is usually very successful. Breathing techniques may also help to facilitate the pelvic floor innervation. Functionally, both muscle groups are closely linked together:

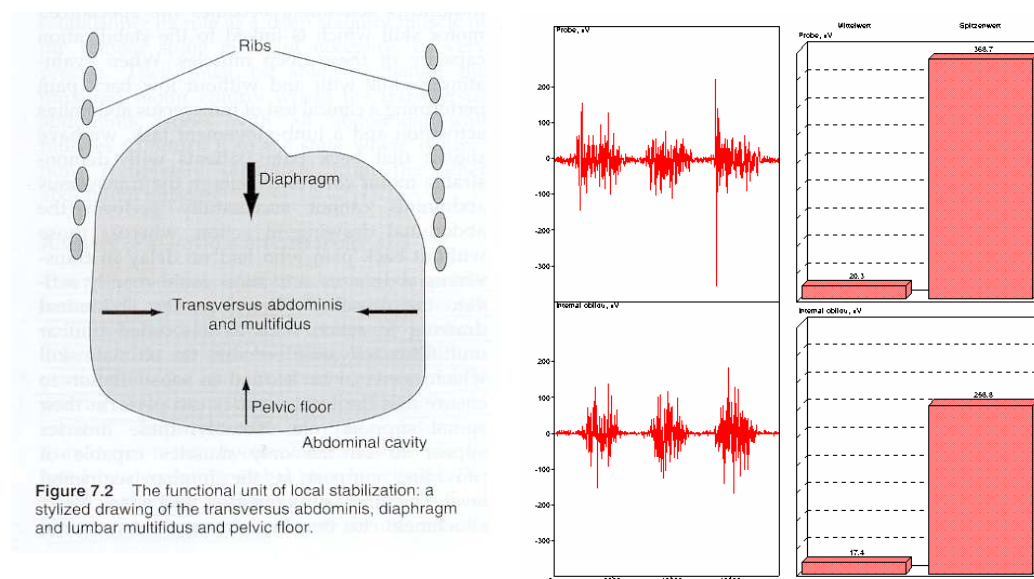
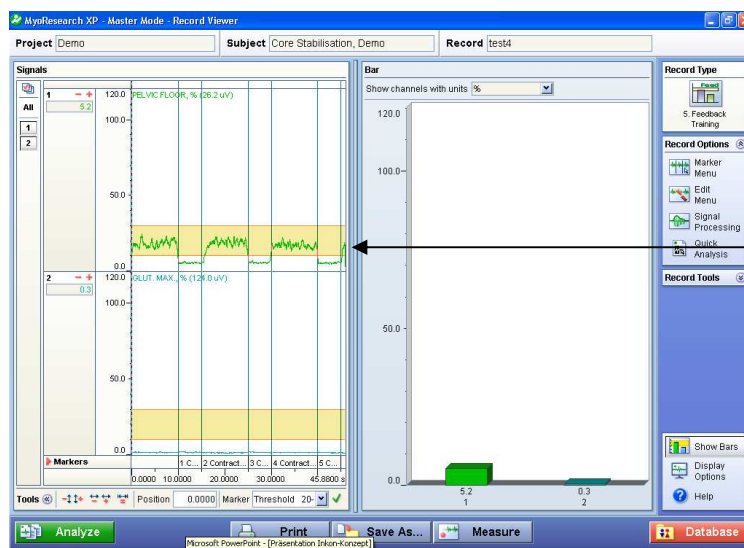


Fig. 9: Schematic drawing (left) of the functional muscle cylinder of deep trunk muscles: EMG raw recordings (right) of the pelvic floor (upper trace) and internal oblique (lower trace) show a fully synchronized innervation pattern in healthy subjects

One very important effect of the EMG based biofeedback-training is that the patient can “see” and control the correct activation of the selected muscles. Once found, the correct innervation can be linked with the corresponding body feeling, which again effectively trains the body awareness (“muscular re-education”). The following Biofeedback screens show an efficient and isolated pelvic floor contraction (upper trace) with absolute innervation silence of the gluteal muscles (lower trace).



Predefined threshold range set to 30 – 50% MVC

Fig. 10: Biofeedback recording with predefined threshold range (yellow area) for MVC normalized EMG contractions. Upper trace: pelvic floor, lower trace: gluteus max.

3 – Stabilization of the functionally adapted muscle innervation

At this stage of the therapy process, the pelvic floor muscle contraction is integrated into whole body tasks and movements. Exercises such as functional gymnastics and “medical training therapy” are performed. A small handheld Biofeedback device can be used to assist in all the exercises.



Fig. 11: Regular leg press training machine facilitates the pelvic floor contraction

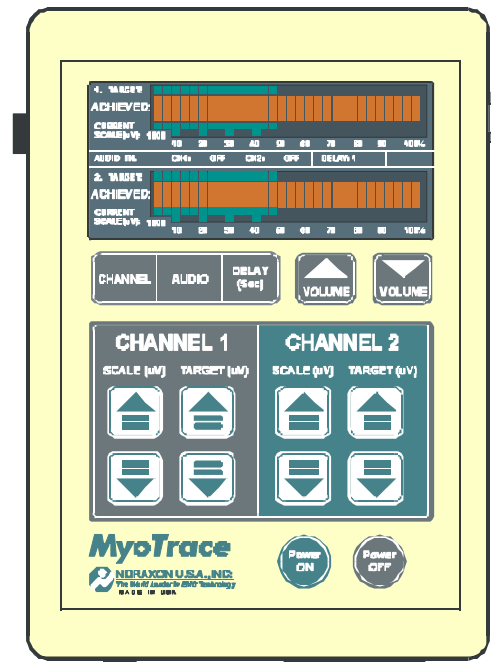


Fig. 12: Two channel biofeedback EMG unit (MyoTrace – Noraxon INC. USA)

The main goal of this stage is to train the pelvic floor contraction during other exercises. This stage is still assisted by isolated muscle training addressing the pelvic floor: Uptraining (increased innervation levels, improved endurance, muscle hypertrophy).



Fig. 13: Telemetric 8 channel EMG recording of trunk and hip muscles with synchronized DV video. The EMG pattern of all involved muscles can be studied while performing regular abdominal training exercise (crunch). Note the high pelvic floor innervation > 50% MVC at peak position.

4 – Integration of the improved pelvic floor muscle innervation

In the last part of the therapy, the improved pelvic floor muscle contraction ability is integrated in general activities of daily living (i.e. coughing, laughing, sneezing, lifting, work and sports). The isolated muscle training strategies concentrates on the maximum activation of the pelvic floor muscle.

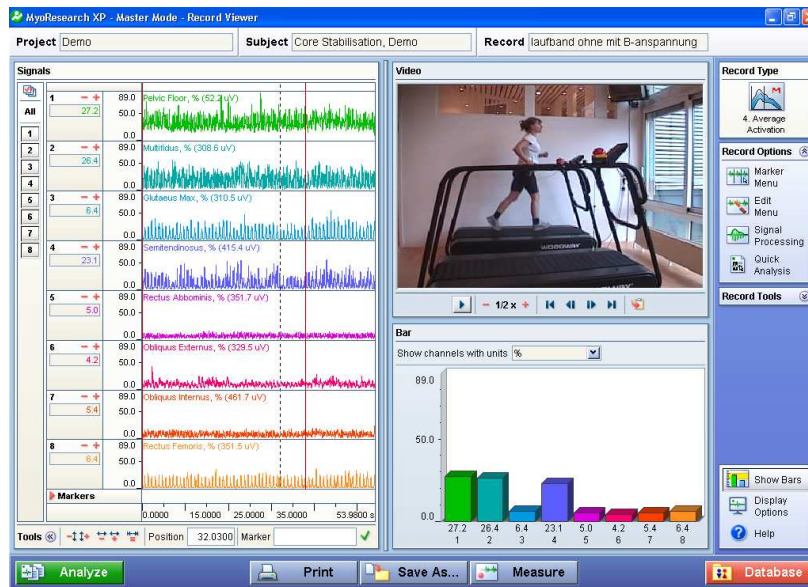


Fig. 14: (Same measurement setup as fig. 13). Treadmill running as a typical daily activity exercise. Note the high contraction level of the pelvic floor muscles (ch. 1)

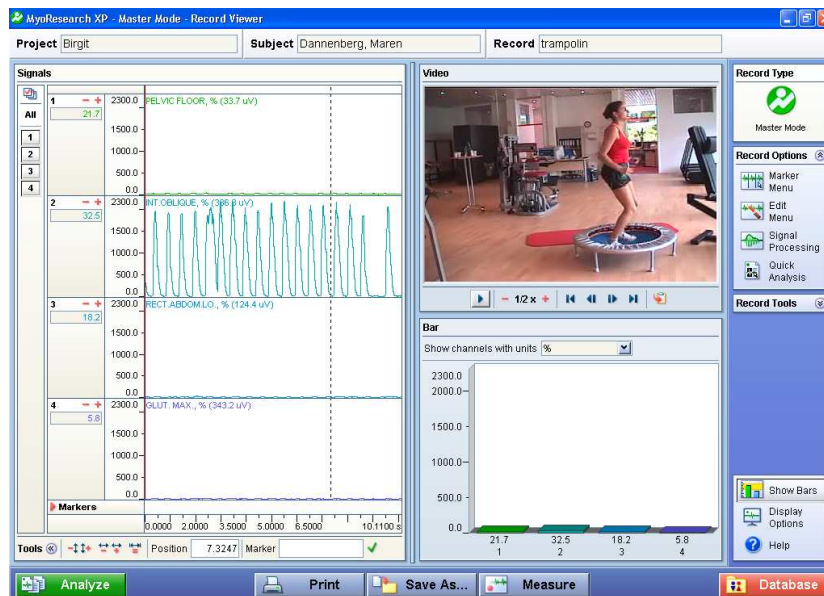


Fig. 15: (Same measurement setup as fig. 13). Hopping on a trampoline.

Retest analysis

After the 6th and 12th therapy sessions, we performed a multi-activity retest. Changes to the maximum innervation level, the innervation constancy, the coordination between synergists and the muscle relaxation ability are compared and documented.

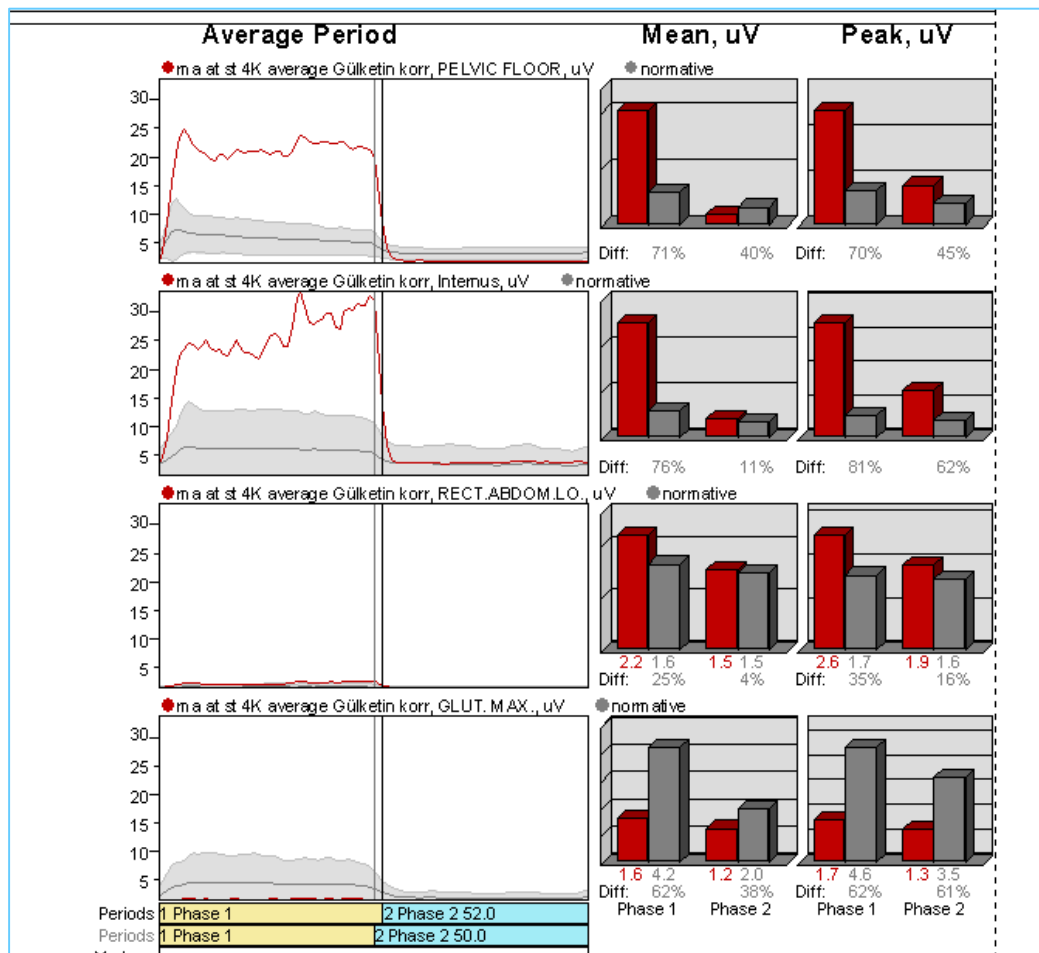
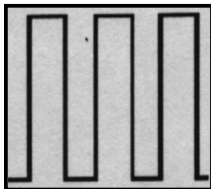


Fig. 16: Test (gray curve) and Retest (red curve) comparison plot. The EMG innervation level of the pelvic floor muscles (ch. 1) and internal oblique (ch. 2) are significantly increased after 12 EMG biofeedback sessions.

Nearly all the patients subjectively report an improvement in their incontinence. This effect is confirmed with an improvement of the SEMG-data of the pelvic floor and the surrounding muscles.

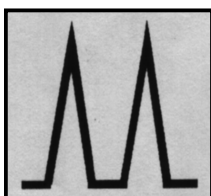
Pelvic Floor Home Training



Isolated Contraction of Pelvic Floor Muscles

Contract pelvic floor muscles while exhaling
Contract without using the gluteal upper abdominal muscles.

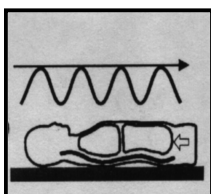
Date	Intensity	Reps.	Sets



Quick Flicks and Relaxation

Contract pelvic floor muscles as quickly as possible and immediately relax again

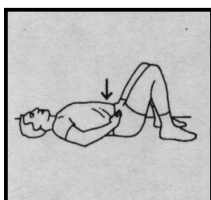
Date	Intensity	Reps.	Sets



Endurance

Keep a constant pelvic floor contraction over several breathing cycles

Date	Intensity	Breathes	Sets



Mobilization, Strengthening and Coordination

Pelvic tilt in prone lying, seated and standing. While exhaling, tilt the pelvis upwards

Date	Intensity	Reps.	Sets